HW #5

1.

(a) $X(z) = \dfrac{1+\frac{1}{2}z^{-1}}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$

$X(z) = \dfrac{z^2+\frac{1}{2}z}{z^2-\frac{3}{4}z+\frac{1}{8}} = \dfrac{z(z+\frac{1}{2})}{(z-\frac{1}{2})(z-\frac{1}{4})} \Rightarrow \dfrac{X(z)}{z} = \dfrac{z+\frac{1}{2}}{(z-\frac{1}{2})(z-\frac{1}{4})} = \dfrac{A_0}{z-\frac{1}{2}} + \dfrac{A_1}{z-\frac{1}{4}}$

$A_0 = (z-\frac{1}{2})\dfrac{X(z)}{z}\Big|_{z=\frac{1}{2}} = \dfrac{z+\frac{1}{2}}{z-\frac{1}{4}}\Big|_{z=\frac{1}{2}} = \dfrac{\frac{1}{2}+\frac{1}{2}}{\frac{1}{2}-\frac{1}{4}} = 4$

$A_1 = (z-\frac{1}{4})\dfrac{X(z)}{z}\Big|_{z=\frac{1}{4}} = \dfrac{z+\frac{1}{2}}{z-\frac{1}{2}}\Big|_{z=\frac{1}{4}} = \dfrac{\frac{1}{4}+\frac{1}{2}}{\frac{1}{4}-\frac{1}{2}} = \dfrac{\frac{3}{4}}{-\frac{1}{4}} = -3$

$X(z) = 4\cdot\dfrac{z}{z-\frac{1}{2}} - 3\cdot\dfrac{z}{z-\frac{1}{4}}$

$x(n) = 4\cdot(\frac{1}{2})^n u(n) - 3\cdot(\frac{1}{4})^n u(n)$

(b) $X(z) = \dfrac{(z+\frac{1}{2})(z+\frac{1}{4})}{(z-\frac{3}{8})(z-\frac{1}{4})}$

$\dfrac{X(z)}{z} = \dfrac{(z+\frac{1}{2})(z+\frac{1}{4})}{z(z-\frac{3}{8})(z-\frac{1}{4})} = \dfrac{A_0}{z} + \dfrac{A_1}{z-\frac{3}{8}} + \dfrac{A_2}{z-\frac{1}{4}}$

$A_0 = z\cdot\dfrac{X(z)}{z}\Big|_{z=0} = \dfrac{(z+\frac{1}{2})(z+\frac{1}{4})}{(z-\frac{3}{8})(z-\frac{1}{4})}\Big|_{z=0} = \dfrac{\frac{1}{2}\cdot\frac{1}{4}}{(-\frac{3}{8})\cdot(-\frac{1}{4})} = \dfrac{4}{3}$

$A_1 = (z-\frac{3}{8})\dfrac{X(z)}{z}\Big|_{z=\frac{3}{8}} = \dfrac{(z+\frac{1}{2})(z+\frac{1}{4})}{z\cdot(z-\frac{1}{4})}\Big|_{z=\frac{3}{8}} = \dfrac{(\frac{3}{8}+\frac{1}{2})(\frac{3}{8}+\frac{1}{4})}{\frac{3}{8}\cdot(\frac{3}{8}-\frac{1}{4})} = \dfrac{\frac{7}{8}\cdot\frac{5}{8}}{\frac{3}{8}\cdot\frac{1}{8}} = \dfrac{35}{3}$

$A_2 = (z-\frac{1}{4})\dfrac{X(z)}{z}\Big|_{z=\frac{1}{4}} = \dfrac{(z+\frac{1}{2})(z+\frac{1}{4})}{z\cdot(z-\frac{3}{8})}\Big|_{z=\frac{1}{4}} = \dfrac{(\frac{1}{4}+\frac{1}{2})(\frac{1}{4}+\frac{1}{4})}{\frac{1}{4}\cdot(\frac{1}{4}-\frac{3}{8})} = \dfrac{\frac{3}{4}\cdot\frac{1}{2}}{\frac{1}{4}\cdot(-\frac{1}{8})} = -12$

$X(z) = \dfrac{4}{3} + \dfrac{35}{3}\cdot\dfrac{z}{z-\frac{3}{8}} - 12\cdot\dfrac{z}{z-\frac{1}{4}}$

$x(n) = \dfrac{4}{3}\delta(n) + \dfrac{35}{3}(\frac{3}{8})^n u(n) - 12\cdot(\frac{1}{4})^n u(n)$

(c) $X(z) = \dfrac{z(z+2)}{z^2+4z+6}$

$\dfrac{X(z)}{z} = \dfrac{z+2}{z^2+4z+6} = \dfrac{z+2}{(z-p_1)(z-p_2)}$ where $p_1 = -2+\sqrt{2}j$ $p_2 = -2-\sqrt{2}j$

$= \dfrac{A_0}{z-p_1} + \dfrac{A_1}{z-p_2}$

$A_0 = (z-p_1)\dfrac{X(z)}{z}\Big|_{z=p_1} = \dfrac{z+2}{z-p_2}\Big|_{z=p_1} = \dfrac{-2+\sqrt{2}j+2}{-2+\sqrt{2}j-(-2-\sqrt{2}j)} = \dfrac{\sqrt{2}j}{2\sqrt{2}j} = \dfrac{1}{2}$

$A_1 = (z-p_2)\dfrac{X(z)}{z}\Big|_{z=p_2} = \dfrac{z+2}{z-p_1}\Big|_{z=p_2} = \dfrac{-2-\sqrt{2}j+2}{-2-\sqrt{2}j-(-2+\sqrt{2}j)} = \dfrac{-\sqrt{2}j}{-2\sqrt{2}j} = \dfrac{1}{2}$

$X(z) = \dfrac{1}{2}\cdot\dfrac{z}{z-p_1} + \dfrac{1}{2}\cdot\dfrac{z}{z-p_2}$

$x(n) = \dfrac{1}{2}(-2+\sqrt{2}j)^n u(n) + \dfrac{1}{2}(-2-\sqrt{2}j)^n u(n)$

(2)

2. (a) $x(n) = \left(\frac{1}{2}\right)^{n-1} u(n-1)$ $\quad X(z) = z^{-1} \cdot \frac{z}{z-\frac{1}{2}} = \frac{1}{z-\frac{1}{2}}$

$h(n) = \left(\frac{1}{3}\right)^{n-1} \cdot u(n) = 3 \cdot \left(\frac{1}{3}\right)^n u(n)$ $\quad H(z) = 3 \cdot \frac{z}{z-\frac{1}{3}}$

$X(z) \cdot H(z) = \frac{1}{z-\frac{1}{2}} \cdot \frac{3z}{z-\frac{1}{3}} = \frac{3z}{(z-\frac{1}{2})(z-\frac{1}{3})}$

$\frac{X(z)H(z)}{z} = \frac{3}{(z-\frac{1}{2})(z-\frac{1}{3})} = \frac{A_0}{z-\frac{1}{2}} + \frac{A_1}{z-\frac{1}{3}}$

$A_0 = (z-\frac{1}{2}) \cdot \frac{X(z)H(z)}{z}\Big/_{z=\frac{1}{2}} = \frac{3}{z-\frac{1}{3}}\Big/_{z=\frac{1}{2}} = \frac{3}{\frac{1}{2}-\frac{1}{3}} = 18$

$A_1 = (z-\frac{1}{3}) \cdot \frac{X(z)H(z)}{z}\Big/_{z=\frac{1}{3}} = \frac{3}{z-\frac{1}{2}}\Big/_{z=\frac{1}{3}} = \frac{3}{\frac{1}{3}-\frac{1}{2}} = -18$

$X(z)H(z) = 18 \cdot \frac{z}{z-\frac{1}{2}} - 18 \cdot \frac{z}{z-\frac{1}{3}}$

$\Rightarrow$ $\boxed{x(n) \textcircled{x} h(n) = 18 \cdot \left(\frac{1}{2}\right)^n u(n) - 18 \cdot \left(\frac{1}{3}\right)^n u(n)}$

(b) $x(n) = [1, -1, 2, -1] \Rightarrow X(z) = 1 - z^{-1} + 2z^{-2} - z^{-3}$

$h(n) = [1, 0, -2, 3] \Rightarrow H(z) = 1 - 2z^{-2} + 3z^{-3}$

$X(z) \cdot H(z) = (1 - z^{-1} + 2z^{-2} - z^{-3})(1 - 2z^{-2} + 3z^{-3}) = 1 - 2z^{-2} + 3z^{-3} - z^{-1} + 2z^{-3} - 3z^{-4} + 2z^{-2} - 4z^{-4} + 6z^{-5} - z^{-3} + 2z^{-5} - 3z^{-6}$

$= 1 - z^{-1} + 4z^{-3} - 7z^{-4} + 8z^{-5} - 3z^{-6}$

$\Rightarrow$ $\boxed{x(n) \textcircled{x} h(n) = [1, -1, 0, 4, -7, 8, -3]}$

3. (a) $x(n) = u(n) \Rightarrow X(z) = \frac{z}{z-1}$

$X(z) \cdot H(z) = \frac{z}{z-1} \cdot \frac{z-2}{z^2+4z+3} = \frac{z(z-2)}{(z-1)(z+1)(z+3)}$

$\frac{X(z) \cdot H(z)}{z} = \frac{z-2}{(z-1)(z+1)(z+3)} = \frac{A_0}{z-1} + \frac{A_1}{z+1} + \frac{A_2}{z+3}$

$A_0 = (z-1) \cdot \frac{X(z)H(z)}{z}\Big/_{z=1} = \frac{z-2}{(z+1)(z+3)}\Big/_{z=1} = \frac{1-2}{(1+1)(1+3)} = -\frac{1}{8}$

$A_1 = (z+1) \cdot \frac{X(z)H(z)}{z}\Big/_{z=-1} = \frac{z-2}{(z-1)(z+3)}\Big/_{z=-1} = \frac{-1-2}{(-1-1)(-1+3)} = \frac{3}{4}$

$A_2 = (z+3) \cdot \frac{X(z)H(z)}{z}\Big/_{z=-3} = \frac{z-2}{(z-1)(z+1)}\Big/_{z=-3} = \frac{-3-2}{(-3-1)(-3+1)} = \frac{-5}{(-4)(-2)} = -\frac{5}{8}$

$$X(z)\cdot H(z) = -\frac{1}{8}\cdot\frac{z}{z-1} + \frac{3}{4}\cdot\frac{z}{z+1} - \frac{5}{8}\cdot\frac{z}{z+3}$$

$$x(n)\otimes h(n) = -\frac{1}{8}u(n) + \frac{3}{4}(-1)^n u(n) - \frac{5}{8}(-3)^n u(n)$$

(b) $$H(z) = \frac{z-2}{z^2+4z+3} = \frac{1-2z^{-2}}{1+4z^{-1}+3z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1+4z^{-1}+3z^{-2}\right] = X(z)\left(1-2z^{-2}\right)$$

$$Y(z) + 4z^{-1}Y(z) + 3z^{-2}Y(z) = X(z) - 2z^{-2}X(z)$$

$$y(n) + 4y(n-1) + 3y(n-2) = x(n) - 2x(n-2)$$

(c) No stable, since, $p_2 = -3$, is not inside the unit circle.